

GATE CSE 2026

Predicted Paper

GATE CS 2026 - Predicted Paper 1: Core Fundamentals

Focus: Algorithms, Theory of Computation, Digital Logic, COA

Total Questions: 65 | Total Marks: 100 | Duration: 180 min

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General Aptitude

Q1. [1 mark] Select the word most OPPOSITE in meaning to: EPHEMERAL

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q2. [1 mark] Average of 5 consecutive odd numbers is 17. Smallest number?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q3. [2 marks] Choose the correct synonym for: PRAGMATIC

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q4. [1 mark] Train 150m long passes a pole in 15 seconds. Speed in km/h?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q5. [2 marks] Sum of Rs 20,000 at 10% p.a. compound interest for 2 years. Amount?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q6. [1 mark] Correct idiom: He cut a _____ figure in the debate.

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q7. [1 mark] If $\log_2(x) = 5$, then $x = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q8. [2 marks] The research findings were _____ by the scientific community, who demanded further evidence.

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q9. [1 mark] Probability of getting sum 7 when two dice are rolled?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q10. [2 marks] Pipes A and B fill a tank in 12 and 18 hours. Together?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Engineering Mathematics

Q11. [1 mark] Determinant of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = ?$

@. [object Object]

A. [object Object]

B. [object Object]

C. [object Object]

Explanation: $\det = 2 \times 2 - 1 \times 1 = 3$.

Q12. [1 mark] $P(A \text{ and } B) = 0.2$, $P(A) = 0.5$, $P(B) = 0.4$. $P(A|B) = ?$

@. [object Object]

A. [object Object]

B. [object Object]

C. [object Object]

Explanation: $P(A|B) = P(A \text{ and } B) / P(B) = 0.2 / 0.4 = 0.5$.

Q13. [1 mark] $d/dx(x^3) = ?$

@. [object Object]

A. [object Object]

B. [object Object]

C. [object Object]

Explanation: Power rule: $d/dx(x^n) = n \cdot x^{(n-1)}$.

Q14. [2 marks] Eigenvalues of $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} = ?$

@. [object Object]

A. [object Object]

B. [object Object]

C. [object Object]

Explanation: Characteristic: $(4 - \lambda)^2 - 1 = 0$. $\lambda^2 - 8\lambda + 15 = 0$. $\lambda = 3, 5$.

Q15. [2 marks] Bayes: $P(B|A) = 0.9$, $P(A) = 0.01$, $P(B|\text{not } A) = 0.1$. $P(A|B) = ?$

@. [object Object]

A. [object Object]

B. [object Object]

C. [object Object]

Explanation: $P(B) = 0.9 \times 0.01 + 0.1 \times 0.99 = 0.108$. $P(A|B) = 0.009 / 0.108 = 0.0833$.

Technical (CSE)

Q16. [1 mark] Best-case time complexity of QuickSort?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Best case when pivot divides array into equal halves at every step.

Q17. [1 mark] Single-source shortest path with non-negative weights?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Dijkstra is optimal for non-negative weights. Bellman-Ford handles negative weights.

Q18. [1 mark] Master Theorem: $T(n)=9T(n/3)+n$ gives complexity?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $a=9, b=3, n^{(\log_3 9)} = n^2$. Since $f(n)=n = O(n^{(2-\epsilon)})$, case 1 applies: $O(n^2)$.

Q19. [2 marks] Time complexity of Strassen matrix multiplication?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Strassen uses 7 multiplications instead of 8. Complexity $O(n^{\log_2 7}) = O(n^{2.81})$.

Q20. [2 marks] LCS time complexity using DP?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: DP table has $(m+1)(n+1)$ entries, each computed in $O(1)$. Total $O(m*n)$.*

Q21. [2 marks] DFS traversal time complexity (adjacency list)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: DFS visits each vertex once and each edge once. With adjacency list: $O(V+E)$.

Q22. [2 marks] Kruskal vs Prim for MST - best for sparse graph?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Kruskal sorts edges: $O(E \log E)$. Prim with binary heap: $O(E \log V)$. For sparse E approx V , both similar. Kruskal preferred.

Q23. [2 marks] If any NP-Complete problem is in P, then?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: All NP problems reduce to NP-Complete. If one NPC is in P, all NP is in P. Hence $P = NP$.

Q24. [1 mark] Which grows fastest as n approaches infinity?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Growth hierarchy: $n \log n < n^2 < 2^n < n!$ (by Stirling approximation).

Q25. [1 mark] Which algorithms use Divide and Conquer?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Merge Sort and Quick Sort divide recursively. Binary Search divides search space. Bubble Sort is iterative.

Q26. [2 marks] Minimum cut in undirected graph computed by?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Max-flow min-cut theorem: Ford-Fulkerson computes max flow which equals min cut capacity.

Q27. [2 marks] With distinct edge weights, MST is?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Cut property guarantees unique MST when all edge weights are distinct.

Q28. [1 mark] Which language is NOT regular?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $\{a^n b^n\}$ requires counting equal a and b , impossible with finite automaton (Pumping Lemma).

Q29. [1 mark] Minimal DFA for strings over $\{0,1\}$ ending in 01 needs?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Need 3 states: $q_0(\text{start})$, $q_1(\text{last}=0)$, $q_2(\text{accept last}=01)$.

Q30. [1 mark] CFG for balanced parentheses?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $S \rightarrow SS$ allows nesting. (S) wraps in parentheses. epsilon allows empty string.

Q31. [2 marks] Language $L = \{a^p \mid p \text{ is prime}\}$ is?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Primality cannot be enforced by CFG. Not regular (Pumping Lemma). TM can check primality so CSL.

Q32. [1 mark] RE equivalent to $(a+b)^*a(a+b)^*$ describes?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

*Explanation: $(a+b)^*a(a+b)^*$ forces at least one a at some position. After first a , anything is allowed.*

Q33. [2 marks] Which problem is UNDECIDABLE?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Halting Problem is classic undecidable. DFA/CFG emptiness are decidable.

Q34. [1 mark] PDA recognizes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: PDAs accept exactly Context-Free Languages, equivalent to languages accepted by CFGs.

Q35. [1 mark] Minimal DFA for $(a+b)^*aa(a+b)^*$ needs?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: States: $q_0(\text{start})$, $q_1(\text{last}=a)$, $q_2(\text{last}=aa, \text{accepting})$. Minimum 3 states.

Q36. [2 marks] Consider CFG: $S \rightarrow aSb \mid \epsilon$. Language $L(G)$ is?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Each production adds one a and one b . After k applications + epsilon: $a^k b^k$.

Q37. [2 marks] Rice Theorem states:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Rice Theorem: any non-trivial property of recursively enumerable languages is undecidable.

Q38. [1 mark] NFA with n states -> equivalent DFA has at most?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Subset construction: DFA states = power set of NFA states -> up to 2^n states.

Q39. [1 mark] 2's complement of -15 (8-bit) is?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $+15 = 00001111$. Invert -> 11110000 . Add 1 -> 11110001 .

Q40. [1 mark] Minimized $F(A,B,C,D) = \sum m(0,1,2,4,5,6,8,9,12,13,14) = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Largest groups on K-map: A prime covers 8 cells, $C'D'$ covers remaining. Result: $A' + C'D'$.

Q41. [1 mark] How many 2-input XOR gates needed for XNOR?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $XNOR = XOR + NOT$. One XOR gate plus one inverter.

Q42. [1 mark] Excess-3 code for decimal 6?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Excess-3 = decimal + 3 in binary. $6 + 3 = 9 \rightarrow 1001$.

Q43. [2 marks] 4:1 MUX with $S_1S_0=10$, $D_0=0$, $D_1=1$, $D_2=A$, $D_3=A'$. Output?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Select 10 means D_2 is chosen. $D_2 = A$.

Q44. [2 marks] JK FF with $J=K=1$, initial $Q=0$. Q after 3 clock pulses?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: JK with $J=K=1$ toggles each pulse. Sequence: $0 \rightarrow 1 \rightarrow 0 \rightarrow 1$. After 3 toggles, $Q=1$.

Q45. [1 mark] Dual of $A + (B.C) = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Dual: swap + and dot (AND/OR). $A + (B.C) \rightarrow A.(B+C)$.

Q46. [2 marks] 4-level AND-OR circuit, each gate $t_{pd}=10ns$. Total propagation delay?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: 4 levels x 10ns per level = 40ns total propagation delay.

Q47. [1 mark] Direct-mapped: 16 blocks cache, 128 blocks MM, 16 words/block. Block offset bits?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Block size = 16 words. Block offset = $\log_2(16) = 4$ bits.

Q48. [2 marks] CPI in ideal 4-stage pipeline (IF, ID, EX, WB)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Ideal pipeline: one instruction completes every cycle after fill. $CPI = 1$.

Q49. [1 mark] ADD R1,(R2) uses which addressing mode?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: (R2) means memory at address in R2. Effective address = contents of R2.

Q50. [2 marks] Opcode is 6 bits. Max distinct operations?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $2^6 = 64$ distinct operations.

Q51. [2 marks] 4-way set-associative, 32 blocks, 4096 MM blocks of 32 bytes. Tag bits?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Offset = $\log_2(32) = 5$. Sets = $32/4 = 8 \rightarrow$ index = 3. MM block addr = 12. Tag = $12 - 3 = 9$. Closest option 10.

Q52. [1 mark] 8-bit signed range in 2's complement?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: n-bit 2's complement: $-2^{(n-1)}$ to $2^{(n-1)}-1$. For n=8: -128 to 127.

Q53. [1 mark] Which memory is fastest?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Cache (SRAM) is fastest, closest to CPU.

Q54. [1 mark] Token, Pattern, Lexeme relation?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Scanner matches Lexeme against Pattern to produce Token.

Q55. [2 marks] SLR parser uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: SLR = LR(0) items + FOLLOW-based reductions. LALR = merged LR(1).

Q56. [2 marks] Backpatching used for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Backpatching fills jump addresses for boolean code and control flow in intermediate code.

Q57. [1 mark] Three-address code for $a=b+c$?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Three-address code uses temporaries: $T1 = b + c$; $a = T1$.

Q58. [1 mark] LR parser is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: LR parsers build rightmost derivation in reverse $\rightarrow$ bottom-up.

Q59. [2 marks] S-attributed definition uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: S-attributed = synthesized only, evaluated in bottom-up (LR) parsing.

Q60. [1 mark] FCFS main drawback?

- @. [object Object]
- A. [object Object]

- B. [object Object]
- C. [object Object]

Explanation: FCFS can cause convoy effect where short processes wait behind long ones.

Q61. [2 marks] Bankers algorithm is for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Banker checks if allocating keeps system in safe state -> deadlock avoidance.

Q62. [1 mark] Page replacement FIFO may show:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Beladys anomaly: more frames can increase page faults in FIFO replacement.

Q63. [2 marks] Semaphore with initial value 1 is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Initial value 1 makes it binary semaphore. Used as mutex for mutual exclusion.

Q64. [2 marks] Effective memory access time: TLB hit=80%, TLB=20ns, MM=100ns. EAT?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $EAT = 0.8 \times 20 + 0.2 \times 120 = 16 + 24 = 40ns$.

Q65. [2 marks] Topological sort requires:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Topological sort only defined for DAG (Directed Acyclic Graph).

Answer Key

General Aptitude

Q1: C
Q2: B
Q3: B
Q4: A
Q5: A
Q6: A
Q7: B
Q8: B
Q9: A
Q10: A

Engineering Mathematics

Q11: B
Q12: B
Q13: A
Q14: A
Q15: A

Technical (CSE)

Q16: B
Q17: B
Q18: B
Q19: B
Q20: B
Q21: B
Q22: A
Q23: A
Q24: D
Q25: A, B, C
Q26: B
Q27: A
Q28: A
Q29: B
Q30: A
Q31: C
Q32: A
Q33: C
Q34: B
Q35: B
Q36: A
Q37: B
Q38: B
Q39: B
Q40: C
Q41: B
Q42: A
Q43: C
Q44: B
Q45: A
Q46: C
Q47: C
Q48: A
Q49: C
Q50: B
Q51: C
Q52: B
Q53: B
Q54: A
Q55: A
Q56: B
Q57: A
Q58: B
Q59: A
Q60: B
Q61: B
Q62: B

Q63: D
Q64: A
Q65: B